CS2011 Foundations of Computing Systems
Test 2

Date: 15th October **Duration:** 50 minutes **Maximum marks:** 25

Instructions

- All questions are compulsory.
- Show the important intermediate steps. An unsupported final answer may not receive full credit.
- Use the conventions of the Hack computer and Hack VM unless a question states otherwise.
- Equivalent correct assembly sequences are acceptable.

Questions

1. Consider the following Hack assembly program. The predefined symbols have their usual meanings, and the first new variable is allocated at RAM address 16.

```
    @count
    M=1
(LOOP)
    @count
    D=M
    @END
    D;JGT
    @LOOP
    0;JMP
(END)
    @END
    0;JMP
```

   (a) During the first pass, which symbol-table entries are created for `LOOP` and `END`? Give their ROM addresses. **[2 marks]**

   (b) During the second pass, which RAM address is allocated to `count`? **[1 marks]**

   (c) Give the 16-bit binary A-instructions generated for the first occurrence of `@count` and for `@END`. **[2 marks]**

   (d) Why does an assembler that supports symbols normally make two passes over the program? **[1 marks]**

2. State the responsibility of each assembler module below. Give a one- or two-line answer for each. **[2 marks]**

   (i) Parser

   (ii) Symbol table

3. Initially, `SP=256`. Trace the following VM commands. Complete the table by giving the value of `SP` and the current top stack value after each command. Recall that `SP` points to the next free stack location. **[4 marks]**

```
push constant 7
push constant 3
sub
```

```
push constant 2
add
neg
```

Command just executed	SP	Top stack value
`push constant 7`		
`push constant 3`		
`sub`		
`push constant 2`		
`add`		
`neg`		

4. Translate the VM command below into Hack assembly language. Assume that `LCL` contains the base address of the local segment. Your translation must push the selected value and leave `SP` pointing to the next free stack location. **[4 marks]**

   `push local 2`

5. The following commands occur inside the VM function `Main.count`:

   ```
   label LOOP
   // other VM commands
   if-goto LOOP
   ```

   Translate `if-goto LOOP` into Hack assembly. The command must remove the top value from the stack and jump when that value is not zero. **[3 marks]**

6. Immediately before translating `call Func 1`, suppose `SP=300`.

   (a) List the five values that the call sequence pushes before transferring control to the called function. **[3 marks]**

   (b) After these five pushes and the standard call setup, what new values are assigned to `ARG` and `LCL`? Show the calculation. **[1 marks]**

7. When a VM function executes `return`, how is its return value passed back to the calling function? State where the value is stored and how the caller's stack pointer is repositioned. **[2 marks]**

End of question paper